

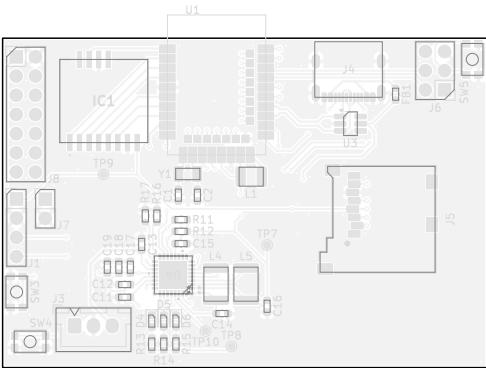
# CanSat Devboard Fabrication Document

## Layer Stack Legend

|  | Material     | Layer         | Thickness        | Dielectric    | Type        | Gerber |
|--|--------------|---------------|------------------|---------------|-------------|--------|
|  |              | F.Paste       |                  |               | Paste Mask  |        |
|  |              | F.Silkscreen  |                  |               | Legend      | GBR    |
|  |              | F.Mask        | 0,02mm           | Solder Resist | Solder Mask | GBR    |
|  | Copper       | L1 (Sig, PWR) | 0,07mm (2,00oz)  |               | Signal      | GBR    |
|  | Prepreg      |               | 0,1mm            | FR4_7628      | Dielectric  |        |
|  | Copper       | L2 (GND)      | 0,035mm (1,00oz) |               | Plane       | GBR    |
|  | Core         |               | 1,21mm           | FR4           | Dielectric  |        |
|  | Copper       | L3 (GND)      | 0,035mm (1,00oz) |               | Plane       | GBR    |
|  | Prepreg      |               | 0,1mm            | FR4_7628      | Dielectric  |        |
|  | Copper       | L4 (Sig, PWR) | 0,07mm (2,00oz)  |               | Signal      | GBR    |
|  | B.Mask       |               | 0,02mm           | Solder Resist | Solder Mask | GBR    |
|  | B.Silkscreen |               |                  |               | Legend      | GBR    |
|  | B.Paste      |               |                  |               | Paste Mask  |        |

Total thickness: 1,66mm  
Note: external layer thicknesses are specified after plating

## Top Fabrication (Scale 1:1)



## Impedance Table

| Transmission Line              | Impedance [ohms] | Tolerance [ohms] | Layer | Trace Width [mm] | Gap [mm] | Ref. Layers |
|--------------------------------|------------------|------------------|-------|------------------|----------|-------------|
| Edge-Coupled Coated Microstrip | 100              | ±10 %            | L1    | 0,2032           | 0,28     | L2          |

### FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge\_Cuts.GBR" SUFFIX.  
  
DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.  
  
SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 4) SURFACE FINISH: IMMERSION GOLD
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLACK.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING YELLOW NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:
  - A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
  - B. Tg 170 °C OR EQUIVALENT.
  - C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY KARPAT SPACE PROGRAM.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

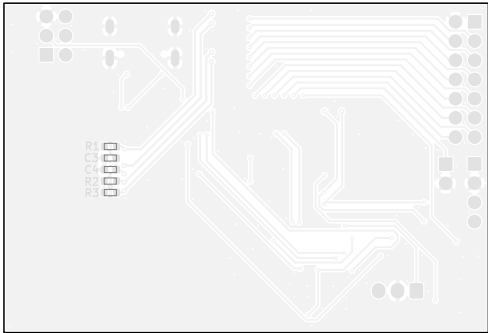
|                  |                    |
|------------------|--------------------|
| BOARD SIZE       | 64,059 × 43,561 mm |
| BOARD THICKNESS  | 1,660 mm           |
| TRACE WIDTH      | 0,200 mm           |
| TRACE TO TRACE   | 0,200 mm           |
| MIN. HOLE (PTH)  | 0,250 mm           |
| MIN. HOLE (NPTH) | 0,650 mm           |
| ANNULAR RING     | 0,150 mm           |
| COPPER TO HOLE   | 0,250 mm           |
| COPPER TO EDGE   | 0,250 mm           |
| HOLE TO HOLE     | 0,100 mm           |
- 11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.
- 12) CONFIRM SPACE WIDTHS AND SPACINGS.

All dimensions are in millimeters unless otherwise specified.

|  |  |                                             |                                         |                            |                                  |                             |
|--|--|---------------------------------------------|-----------------------------------------|----------------------------|----------------------------------|-----------------------------|
|  |  | Comments:                                   | Company:<br>Karpat Space Program        |                            | Variant:<br>PRELIMINARY          | Git Hash:<br>9b0fd68        |
|  |  |                                             | Board Name:<br>CanSat Devboard          |                            | Project Name:<br>Cansat Devboard |                             |
|  |  | Sheet Title:<br>Top Fabrication (Scale 1:1) | File Name:<br>CanSat_Devboard.kicad_pcb | Designer:<br>Bertalan Nagy | Date:<br>2024-04-13              | Revision:<br>+ (Unreleased) |
|  |  | Sheet Path:                                 |                                         | Reviewer:                  | Size:<br>A4                      | Sheet:<br>1 of 10           |

# CanSat Devboard Fabrication Document

## Bottom Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

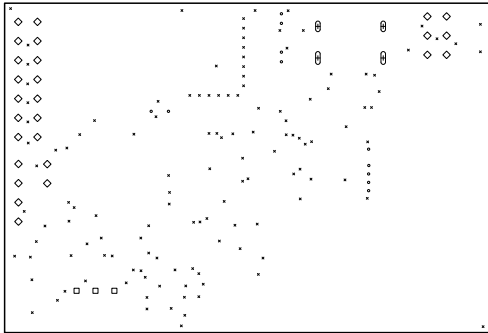
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|  |                                                | Board Name:<br>CanSat Devboard          |                            | Project Name:<br>Cansat Devboard |                             |
|  | Sheet Title:<br>Bottom Fabrication (Scale 1:1) | File Name:<br>CanSat_Devboard.kicad_pcb | Designer:<br>Bertalan Nagy | Date:<br>2024-04-13              | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                                    |                                         | Reviewer:                  | Size:<br>A4                      | Sheet:<br>2 of 10           |

# CanSat Devboard Fabrication Document

## Drill Table

| Symbol    | Count | Hole Size          | Plated | Hole Shape | Drill Layer Pair              | Hole Type |
|-----------|-------|--------------------|--------|------------|-------------------------------|-----------|
| ×         | 130   | 0,25mm (9,84mils)  | PTH    | Round      | L1 (Sig, PWR) - L4 (Sig, PWR) | Via       |
| ○         | 11    | 0,30mm (11,81mils) | PTH    | Round      | L1 (Sig, PWR) - L4 (Sig, PWR) | Via       |
| +         | 4     | 0,60mm (23,62mils) | PTH    | Slot       | L1 (Sig, PWR) - L4 (Sig, PWR) | Pad       |
| □         | 3     | 0,95mm (37,40mils) | PTH    | Round      | L1 (Sig, PWR) - L4 (Sig, PWR) | Pad       |
| ◇         | 26    | 1,00mm (39,37mils) | PTH    | Round      | L1 (Sig, PWR) - L4 (Sig, PWR) | Pad       |
| Total 174 |       |                    |        |            |                               |           |

## Drill Drawing L1 - L4 (Scale 1:1)



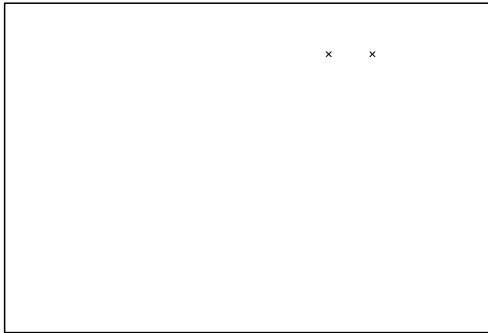
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|  |                                         | Board Name:<br>CanSat Devboard          |                            | Project Name:<br>Cansat Devboard |                             |
|  | Sheet Title:<br>Drill Drawing (L1 - L4) | File Name:<br>CanSat_Devboard.kicad_pcb | Designer:<br>Bertalan Nagy | Date:<br>2024-04-13              | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                             |                                         | Reviewer:                  | Size:<br>A4                      | Sheet:<br>3 of 10           |

# CanSat Devboard Fabrication Document

Drill Table

| Symbol | Count   | Hole Size          | Plated | Hole Shape | Drill Layer Pair              | Hole Type  |
|--------|---------|--------------------|--------|------------|-------------------------------|------------|
| X      | 2       | 0,65mm (25,59mils) | NPTH   | Round      | L1 (Sig, PWR) - L4 (Sig, PWR) | Mechanical |
|        | Total 2 |                    |        |            |                               |            |

Drill Drawing L1 - L4 (Scale 1:1)



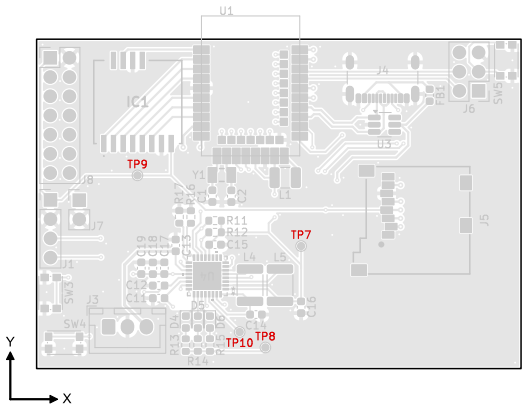
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|  | Comments:                               | Company:<br>Karpát Space Program        |                            | Variant:<br>PRELIMINARY          | Git Hash:<br>9b0fd68        |
|  |                                         | Board Name:<br>CanSat Devboard          |                            | Project Name:<br>Cansat Devboard |                             |
|  | Sheet Title:<br>Drill Drawing (L1 - L4) | File Name:<br>CanSat_Devboard.kicad_pcb | Designer:<br>Bertalan Nagy | Date:<br>2024-04-13              | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                             |                                         | Reviewer:                  | Size:<br>A4                      | Sheet:<br>4 of 10           |

# CanSat Devboard Fabrication Document

Top Test Points (Scale 1:1)

| Ref. | Net   | X [mm] | Y [mm] |
|------|-------|--------|--------|
| TP7  | VMCU  | 87.96  | 1.25   |
| TP8  | VSYS  | 83.23  | -12.16 |
| TP9  | +3.3V | 66.31  | 10.60  |
| TP10 | VLDO  | 79.83  | -10.12 |

| Ref. | Net | X [mm] | Y [mm] |
|------|-----|--------|--------|
|------|-----|--------|--------|



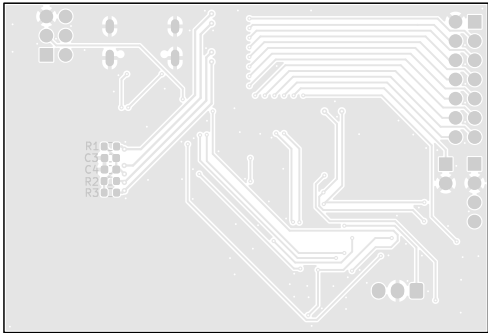
All dimensions are in millimeters unless otherwise specified.

|  |                                             |                                         |                            |                                  |                             |
|--|---------------------------------------------|-----------------------------------------|----------------------------|----------------------------------|-----------------------------|
|  | Comments:                                   | Company:<br>Karpát Space Program        |                            | Variant:<br>PRELIMINARY          | Git Hash:<br>9b0fd68        |
|  |                                             | Board Name:<br>CanSat Devboard          |                            | Project Name:<br>Cansat Devboard |                             |
|  | Sheet Title:<br>Top Test Points (Scale 1:1) | File Name:<br>CanSat_Devboard.kicad_pcb | Designer:<br>Bertalan Nagy | Date:<br>2024-04-13              | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                                 |                                         | Reviewer:                  | Size:<br>A4                      | Sheet:<br>5 of 10           |

# CanSat Devboard Fabrication Document

Bottom Test Points (Scale 1:1)

| Ref. | Net | X [mm] | Y [mm] |
|------|-----|--------|--------|
|------|-----|--------|--------|

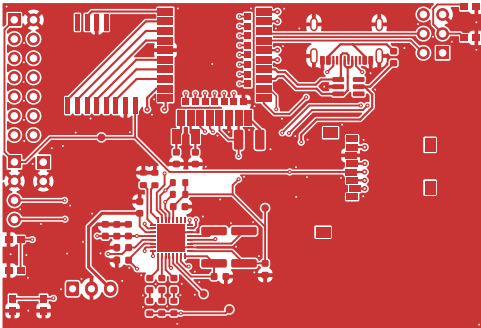


All dimensions are in millimeters unless otherwise specified.

|  |                                                |                                         |                            |                                  |                             |
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|  | Comments:                                      | Company:<br>Karpát Space Program        |                            | Variant:<br>PRELIMINARY          | Git Hash:<br>9b0fd68        |
|  |                                                | Board Name:<br>CanSat Devboard          |                            | Project Name:<br>Cansat Devboard |                             |
|  | Sheet Title:<br>Bottom Test Points (Scale 1:1) | File Name:<br>CanSat_Devboard.kicad_pcb | Designer:<br>Bertalan Nagy | Date:<br>2024-04-13              | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                                    |                                         | Reviewer:                  | Size:<br>A4                      | Sheet:<br>6 of 10           |

# CanSat Devboard Fabrication Document

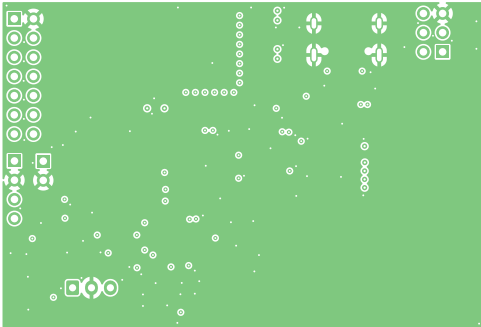
L1 (Sig, PWR) (Scale 1:1)



|  |                                           |                                         |                            |                                  |                             |
|--|-------------------------------------------|-----------------------------------------|----------------------------|----------------------------------|-----------------------------|
|  | Comments:                                 | Company:<br>Karpát Space Program        |                            | Variant:<br>PRELIMINARY          | Git Hash:<br>9b0fd68        |
|  |                                           | Board Name:<br>CanSat Devboard          |                            | Project Name:<br>Cansat Devboard |                             |
|  | Sheet Title:<br>L1 (Sig, PWR) (Scale 1:1) | File Name:<br>CanSat_Devboard.kicad_pcb | Designer:<br>Bertalan Nagy | Date:<br>2024-04-13              | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                               |                                         | Reviewer:                  | Size:<br>A4                      | Sheet:<br>7 of 10           |

# CanSat Devboard Fabrication Document

L2 (GND) (Scale 1:1)



|  |                                                      |                                                         |                                            |                                                  |                                             |
|--|------------------------------------------------------|---------------------------------------------------------|--------------------------------------------|--------------------------------------------------|---------------------------------------------|
|  | <a href="#">Comments:</a>                            | <a href="#">Company:</a><br>Karpát Space Program        |                                            | <a href="#">Variant:</a><br>PRELIMINARY          | <a href="#">Git Hash:</a><br>9b0fd68        |
|  |                                                      | <a href="#">Board Name:</a><br>CanSat Devboard          |                                            | <a href="#">Project Name:</a><br>Cansat Devboard |                                             |
|  | <a href="#">Sheet Title:</a><br>L2 (GND) (Scale 1:1) | <a href="#">File Name:</a><br>CanSat_Devboard.kicad_pcb | <a href="#">Designer:</a><br>Bertalan Nagy | <a href="#">Date:</a><br>2024-04-13              | <a href="#">Revision:</a><br>+ (Unreleased) |
|  | <a href="#">Sheet Path:</a>                          |                                                         | <a href="#">Reviewer:</a>                  | <a href="#">Size:</a><br>A4                      | <a href="#">Sheet:</a><br>8 of 10           |



# CanSat Devboard Fabrication Document

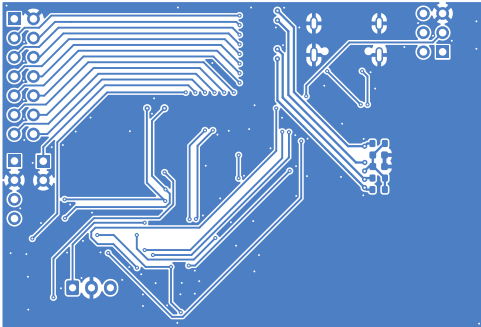
L3 (GND) (Scale 1:1)



|  |                                      |                                         |                            |                                  |                             |
|--|--------------------------------------|-----------------------------------------|----------------------------|----------------------------------|-----------------------------|
|  | Comments:                            | Company:<br>Karpát Space Program        |                            | Variant:<br>PRELIMINARY          | Git Hash:<br>9b0fd68        |
|  |                                      | Board Name:<br>CanSat Devboard          |                            | Project Name:<br>Cansat Devboard |                             |
|  | Sheet Title:<br>L3 (GND) (Scale 1:1) | File Name:<br>CanSat_Devboard.kicad_pcb | Designer:<br>Bertalan Nagy | Date:<br>2024-04-13              | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                          |                                         | Reviewer:                  | Size:<br>A4                      | Sheet:<br>9 of 10           |

# CanSat Devboard Fabrication Document

L4 (Sig, PWR) (Scale 1:1)



|  |                                           |                                         |                            |                                  |                             |
|--|-------------------------------------------|-----------------------------------------|----------------------------|----------------------------------|-----------------------------|
|  | Comments:                                 | Company:<br>Karpát Space Program        |                            | Variant:<br>PRELIMINARY          | Git Hash:<br>9b0fd68        |
|  |                                           | Board Name:<br>CanSat Devboard          |                            | Project Name:<br>Cansat Devboard |                             |
|  | Sheet Title:<br>L4 (Sig, PWR) (Scale 1:1) | File Name:<br>CanSat_Devboard.kicad_pcb | Designer:<br>Bertalan Nagy | Date:<br>2024-04-13              | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                               |                                         | Reviewer:                  | Size:<br>A4                      | Sheet:<br>10 of 10          |